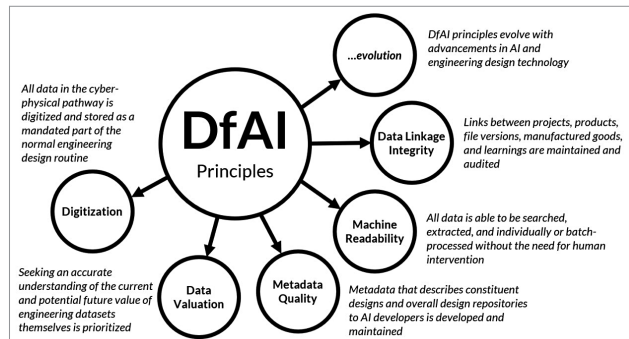




## Design for AI engineering framework developed

Researchers from Carnegie Mellon University have developed a Design for Artificial Intelligence (DfAI) framework in collaboration with researchers at Penn State University



*DfAI Principles (Credit: Carnegie Mellon University, College of Engineering)*

to educate and encourage the academic and industrial engineering community to adopt AI engineering design. The team outlines three personas as necessities for DfAI: engineering designers, design repository curators, and AI developers. An engineering designer may be a person or team responsible for developing the specifications of a new project. They are the problem solvers who can understand the engineering constraints as well as the AI algorithms. Design repository curators must take the role of a database maintainer one step further by having engineering design and manufacturing knowledge to deliver design engineers the data management tools to meet workflow demands and be extensible to future demands. Finally, the AI developers must be able to ideate, develop, market, and continuously improve AI software products to help the design engineers.

## A watch that works only when attached organism is healthy

Researchers at University of Chicago wondered if they could change that fickle relationship by bringing devices to

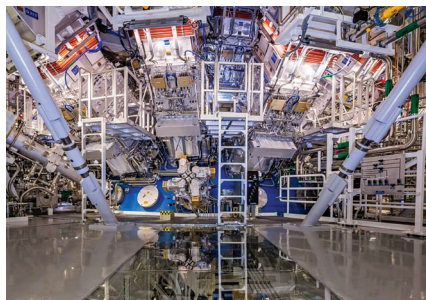


*UChicago researchers have created a watch powered by slime mold—which has to be fed and cared for to work (Credit: HCIntegration)*

life—literally. They used the electrically conductive single-cell organism known as “slime mold” to create a watch that works only when the organism is healthy, requiring the user to provide it with food and care. The organism is placed in an enclosure on the watch, and the user must regularly feed it a mixture of water and oats to induce its growth. When it reaches the other side of the enclosure, it forms an electrical circuit that activates the heart rate monitor function. When tested, researchers found a high level of attachment to the watch, with some users saying it felt like a pet—even naming it or putting their partner in charge of the feeding when they got sick. This also depicted the users’ emotional response when they were asked to neglect the organism, expressing guilt or even grief.

## Fusion reaction generates more energy than used to create it

A team of researchers at the National Ignition Facility (NIF) at Lawrence Livermore National Laboratory have achieved success in producing more energy than the laser pulse that



*Fusion energy creation (Credit: Imperial College London)*

was used to heat the fuel. The NIF focused on inertial confinement fusion, which uses a system of lasers to heat up fuel pellets producing a plasma—a cloud of charged ions. The fuel

pellets contain “heavy” versions of hydrogen—deuterium and tritium—that are easier to fuse and produce more energy. However, the fuel pellets need to be heated and pressurized to conditions found at the centre of the Sun, which is a natural fusion reactor. Once these conditions are achieved, fusion reactions release several particles, including alpha particles, which interact with the surrounding plasma and heat it up further. The heated plasma then releases more alpha particles and so on, in a self-sustaining reaction—a process referred to as ignition.

## New technique found to simplify qubits manufacture

Scientists from Japan have discovered how a superconducting material, niobium nitride, can be added as a flat, crystalline layer to a nitride-semiconductor substrate. This technique could make it simple to manufacture quantum qubits that can be used with conventional computer devices. Niobium nitride can become superconducting at